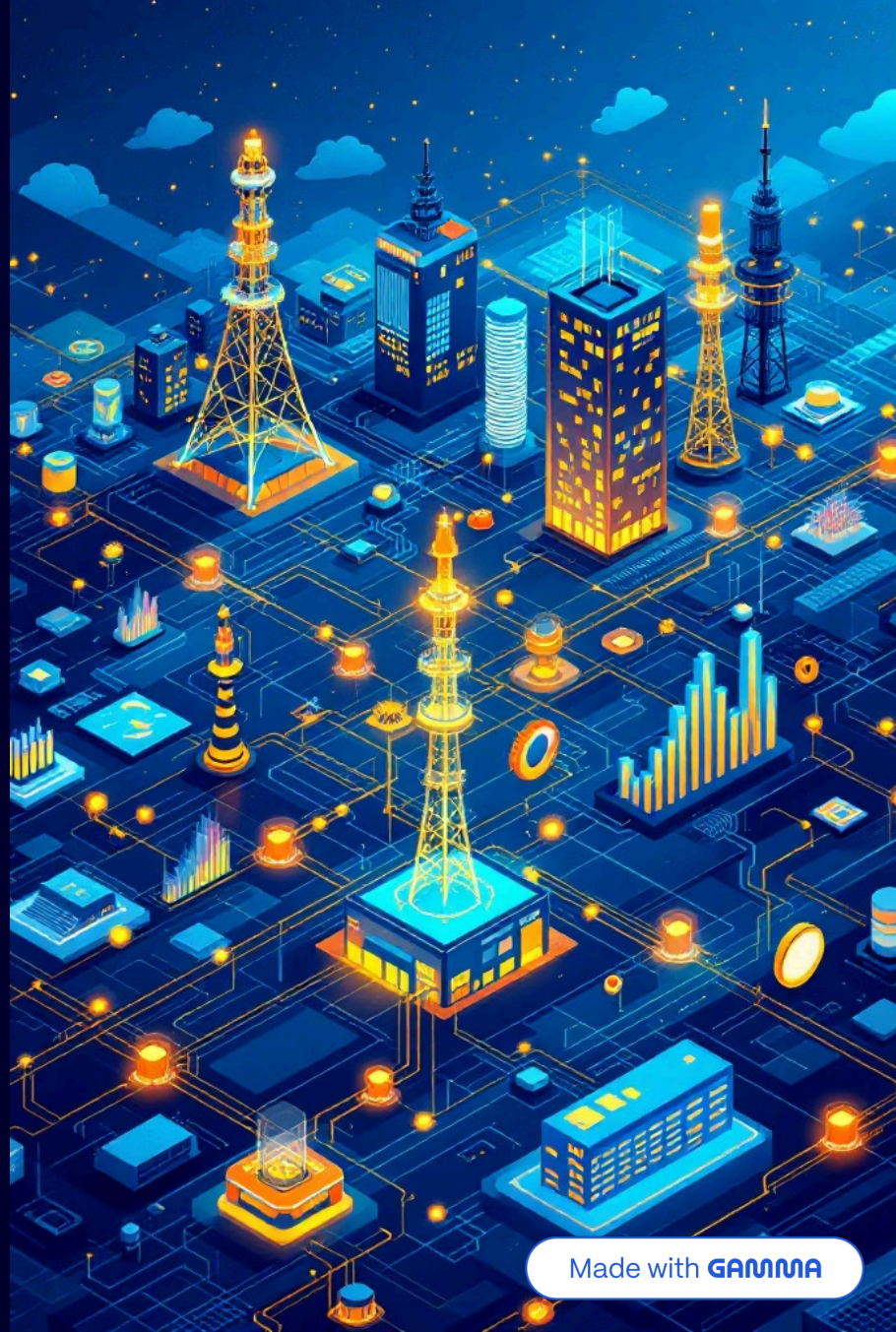
Project Overview: Strategic Churn Analysis & AI Prediction

Optimizing Retention through Data Engineering and Machine Learning

Author: Prateek Bag | **Date:** March 10, 2026



Made with GAMMA

Executive Summary

6,418

Total Customers

Base analyzed across demographics and service types

26.99%

Churn Rate

1,732 customers churned overall

761

Competition-Driven

Primary churn driver (approx. 44%)

Month-to-Month contracts show 47% churn rate. Jammu & Kashmir (57%) and Assam (38%) exhibit highest regional attrition.

Made with GAMMA

Data Engineering Workflow

SQL Backend Architecture

Extensive SQL processing for data cleaning and model preparation.

- Handled NULL values with ISNULL, replacing missing entries with "None" for categorical fields and "No" for service flags
- Created vw_churn view for model training with stayed/churned status
- Developed vw_joined view for future predictions on new joiners
- Built queries calculating churn percentages by gender, state, and contract type

Made with GAMMA

Executive & Demographic Dashboard Insights

Contract Influence

Month-to-Month contracts most volatile. Two Year contracts show ~2% churn rate.

Service Impact

Fiber Optic: 58% churn. Cable: 36%. DSL: 32%.

Payment Behavior

Mailed Check: 38% churn. Bank Withdrawal: 34%. Credit Cards lower.

Age Profile

36-50 age group highest volume: 621 customers churned.

Tenure Correlation

Highest churn within first 6 months of tenure.

Social Status

Non-married customers show higher churn propensity.

Made with GAMMA

Predictive Analytics: Random Forest Classifier

Model Performance

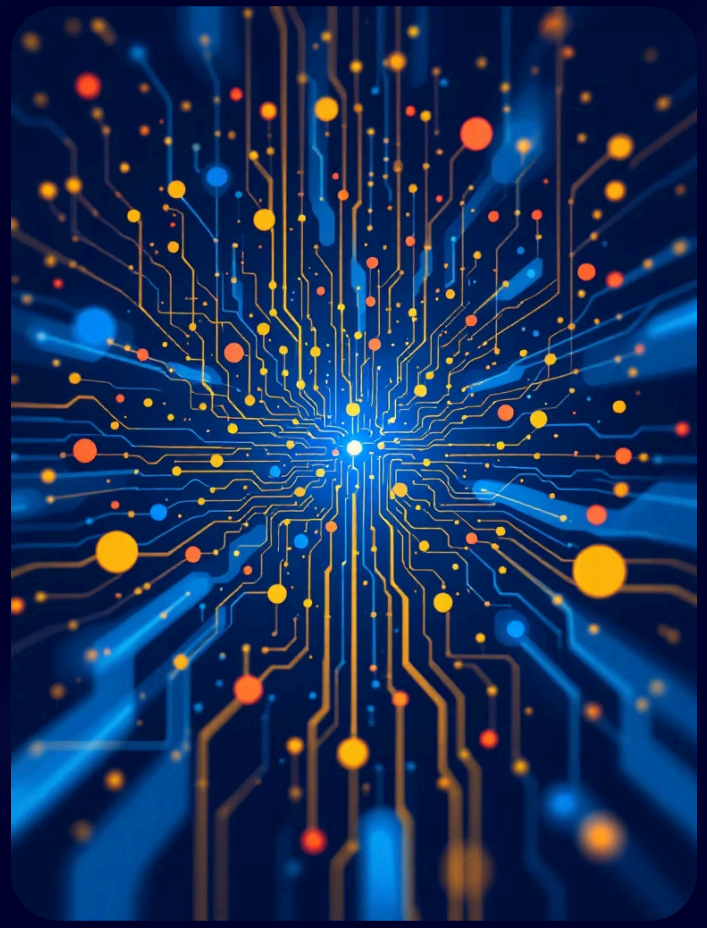
Python implementation achieved 84% accuracy with 0.78 precision for churned class.

Key Features

Contract, Tenure_in_Months, and Monthly_Charges identified as most significant predictors.

Future Risk

Applied model to 411 new joiners, flagging high churn probability customers.



Made with GAMMA

Strategic Recommendations

Fiber Optic Stability

Investigate 58% churn rate in Fiber Optic services. Ensure network stability and competitive pricing.

Contract Migration

Incentivize Month-to-Month customers to move to One Year or Two Year contracts through discounts.

Regional Retention

Launch targeted loyalty programs in Jammu & Kashmir and Assam to combat high regional attrition.

Made with GAMMA

Future Roadmap

Proactive Engagement

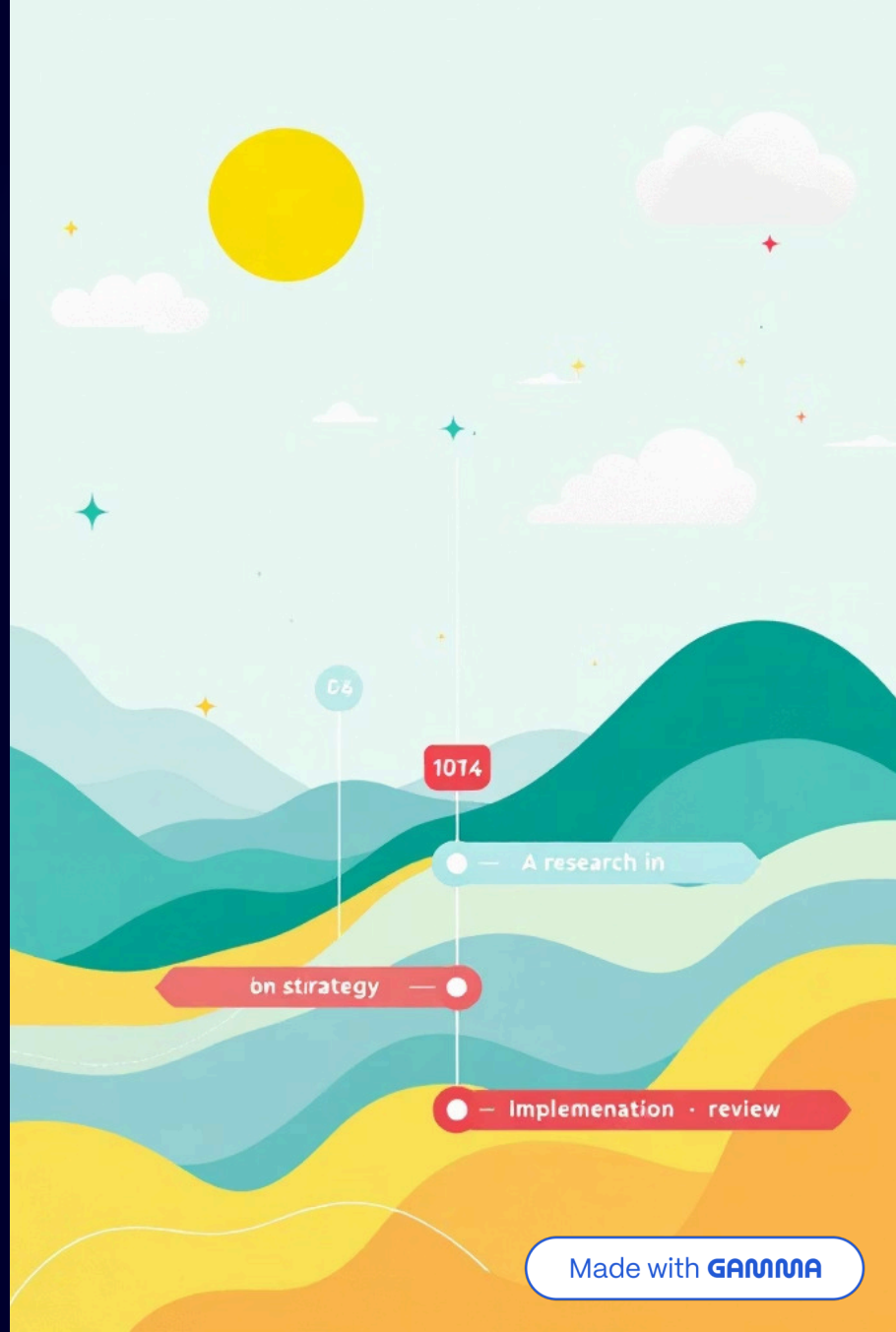
Deploy targeted retention campaigns for high-risk customers identified by predictive model.

Revenue Protection

Implement contract incentives and service improvements to reduce attrition.

Continuous Optimization

Monitor model performance and refine strategies based on regional and demographic insights.



Made with GAMMA